

Programming Fundamentals 2

Topic 04 - Tutorial Sheet 4.2

Static Members, Javadoc

1. Instance vs Static (Class) Variables

Instance Variables

- Belong to **individual objects**
- Each object has its **own copy**
- Represent object-specific state

```
public class BouncingBall {  
    private int xPosition;  
    private int yPosition;  
}
```

If three `BouncingBall` objects exist, there are three separate `xPosition` and `yPosition` values.

Check Your Understanding

1. How many copies of a static variable exist in a program?
 2. Why is `gravity` (as in the Bouncing BALL example) suitable as a static variable, but `xPosition` is not?
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2. Static Methods

Think About This

Should the following method be static or instance-based?

```
public int moveBallDown() { ... }
```

Explain your reasoning.

3. Javadoc: Documenting Classes and Methods

Class Documentation Example

```
/**
 * Represents an employee in the company.
 * Stores salary information and provides
 * methods for salary calculations.
 *
 * @author Your Name
 * @version 1.0
 */
public class Employee {
    ...
}
```

Method Documentation Example

```
/**
 * Calculates and returns the employee's net salary.
 *
 * @return the net salary after deductions
 */
public double calculateNetSalary() {
    return salary - deductions;
}
```

Common Javadoc Tags

- `@author`
- `@version`
- `@param`
- `@return`

Task

Add full Javadoc comments to:

- One class
- One constructor
- Two methods

Hint :

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- if in doubt, check with an API e.g. of Strings or Collections to see the level of comment.
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